23/05/2026

Code-A

Aakash
Medical | IIT-JEE | Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18, Udyog Vihar, Gurugram, Haryana - 122015, ***Ph.***+91-1244168300

MM : 720 Project Phoenix for Re-NEET-2026_PST-04P **Time : 180 Min.**

PHYSICS

1. (4)
2. (3)
3. (2)
4. (4)
5. (4)
6. (4)
7. (2)
8. (4)
9. (3)
10. (1)
11. (4)
12. (2)
13. (2)
14. (2)
15. (2)
16. (3)
17. (2)
18. (4)
19. (1)
20. (2)
21. (2)
22. (2)
23. (1)
24. (2)
25. (3)
26. (1)
27. (3)
28. (4)
29. (2)
30. (2)
31. (4)
32. (4)
33. (4)
34. (2)
35. (1)
36. (3)
37. (3)
38. (2)
39. (1)
40. (4)
41. (4)
42. (2)
43. (4)
44. (4)
45. (2)

CHEMISTRY

46. (3)
69. (3)

47. (2)
48. (3)
49. (3)
50. (4)
51. (3)
52. (4)
53. (3)
54. (1)
55. (2)
56. (3)
57. (2)
58. (1)
59. (4)
60. (2)
61. (3)
62. (2)
63. (2)
64. (3)
65. (2)
66. (2)
67. (1)
68. (1)
70. (4)
71. (3)
72. (2)
73. (3)
74. (3)
75. (4)
76. (1)
77. (2)
78. (1)
79. (4)
80. (3)
81. (2)
82. (1)
83. (4)
84. (1)
85. (1)
86. (3)
87. (1)
88. (2)
89. (1)
90. (3)

Aakash Medical|IIT-JEE|Foundations

BOTANY

91. (4)
92. (2)
93. (2)
94. (4)
95. (2)
96. (2)
97. (4)
98. (1)
99. (4)
100. (4)
101. (4)
114. (3)
115. (2)
116. (1)
117. (2)
118. (3)
119. (1)
120. (2)
121. (3)
122. (1)
123. (3)
124. (2)

102. (2)
103. (3)
104. (1)
105. (4)
106. (4)
107. (2)
108. (1)
109. (3)
110. (2)
111. (2)
112. (4)
113. (2)
125. (2)
126. (2)
127. (4)
128. (3)
129. (1)
130. (3)
131. (4)
132. (4)
133. (1)
134. (4)
135. (2)

ZOOLOGY

136. (3)
137. (1)
138. (3)
139. (2)
140. (4)
141. (3)
142. (4)
143. (2)
144. (3)
145. (3)
146. (4)
147. (3)
148. (4)
149. (4)
150. (3)
151. (2)
152. (1)
153. (3)
154. (4)
155. (2)
156. (3)
159. (3)
160. (2)
161. (1)
162. (2)
163. (1)
164. (3)
165. (2)
166. (2)
167. (4)
168. (2)
169. (3)
170. (1)
171. (1)
172. (4)
173. (4)
174. (4)
175. (3)
176. (3)
177. (1)
178. (3)
179. (1)

Aakash Medical|IIT-JEE|Foundations

157. (3)

158. (2)

180. (3)

Aakash Medical|IIT-JEE|Foundations

Hints and Solutions

PHYSICS

(1) **Answer :** (4)
Solution:
The cylindrical lens is used to correct astigmatism.

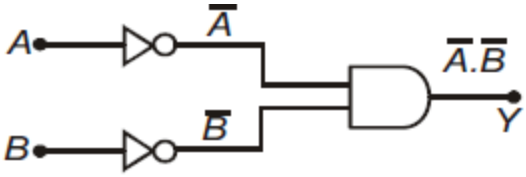
(2) **Answer :** (3)
Solution:
A plane mirror can form virtual image of real object. The image formed will be erect and of same size.

(3) **Answer :** (2)
Solution:
The wavelength of Balmer series of hydrogen atom mostly lies in visible region.

(4) **Answer :** (4)
Solution:
$v = \frac{1}{\sqrt{\varepsilon_0 \varepsilon_r \mu_0 \mu_r}}$

(5) **Answer :** (4)
Solution:
Energy of photon, $E = hf$ ⇒ A is correct
Photons are neutral ⇒ B is correct
The rest mass of photon is zero ⇒ C is correct

(6) **Answer :** (4)
Solution:



$Y = \overline{A} \,.\, \overline{B}$
$\overline{A + B} = \overline{A} \,.\, \overline{B}$

(7) **Answer :** (2)
Solution:
$^{237}_{93}Y \rightarrow {}^{229}_{89}Z + 2\,^{4}_{2}\text{He} + \text{Energy}$
$^{A}_{Z}X \rightarrow {}^{237}_{93}Y + 2\,^{0}_{+1}\beta + 2\nu$
Z = 95; A = 237

(8) **Answer :** (4)
Solution:
Here k = 15, ω = 3×10^8
And $v = \frac{\omega}{k} = \frac{3\times10^8}{15} = 2 \times 10^7$ m/s

(9) **Answer :** (3)
Solution:
Angular width = $\frac{2\lambda}{d}$
$= \frac{2\times6000\times10^{-10}}{12\times10^{-5}\times10^{-2}}$
= 1 rad

(10) **Answer :** (1)
Solution:

Aakash Medical|IIT-JEE|Foundations

$\delta \text{ min } = 2i - A$

$i = \frac{\delta \text{ min } + A}{2} = \frac{30+60}{2} = 45^{\circ}$

(11) Answer : (4)

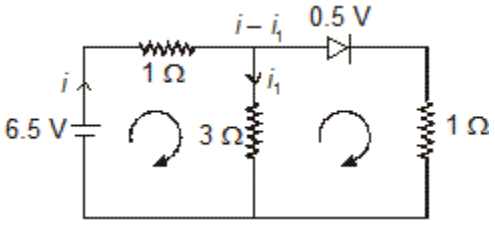
Solution:

For the condition of dispersion without deviation

$\frac{A'}{A} = \frac{\mu-1}{\mu'-1} \Rightarrow \frac{5}{A} = \frac{1.5-1}{1.4-1} \Rightarrow \frac{5}{A} = \frac{0.5}{0.4} \Rightarrow A = 4^{\circ}$

(12) Answer : (2)

Solution:



$3i_1 + i = 6.5$ ... (i)

$4i_1 - i = 0.5$ ... (ii)

On solving (i) and (ii), we get

$i_1 = 1$ A

$i - i_1 = 2.5$ A

Thus,

$P = v(i - i_1) = 0.5 \times 2.5 = 1.25$ watt

(13) Answer : (2)

Solution:

$F_{\alpha} = \frac{R}{\mu-1}$

$= \frac{10}{1.5-1} = 20$

$F_e = 20$ cm

$F_m = \infty$

$\frac{1}{F} = \frac{2}{|F_u|} + \frac{1}{|F_m}$

$F = 10$ cm

On silvering convex lens behave like concave mirror

Hence $F = -10$ cm

Mirror formula

$\frac{1}{F} = \frac{1}{v} + \frac{1}{v}$

$-\frac{1}{10} = \frac{1}{u} - \frac{1}{20}$

$\frac{1}{v} = -\frac{1}{10} + \frac{1}{20}$

v = – 20 cm

(14) Answer : (2)

Solution:

$n = \frac{6.4}{200 \times 1.6 \times 10^{-13}} = 2 \times 10^{11}$

(15) Answer : (2)

Solution:

Number of neutrons = Mass number – atomic number = 24 – 11 = 13

(16) Answer : (3)

Solution:

By Einstein's equation

$ev_0 = hf - \varphi$

⇒ If f will double, then stopping potential will become more than doubled.

(17) Answer : (2)

Solution:

$\lambda_L = \frac{1}{R}$

Aakash Medical|IIT-JEE|Foundations

$\lambda_p = \frac{9}{R}$

$\frac{\lambda_L}{\lambda_P} = \frac{R}{R \times 9}$

(18) Answer : (4)
Solution:
For concave mirror, minimum distance between a real object and its real image is zero, when object is placed at centre of curvature.

(19) Answer : (1)
Solution:

$\because \quad \lambda = \frac{h}{p}$

$= \frac{h}{\sqrt{2mE}}$

$\lambda \alpha \frac{1}{\sqrt{E}}$

$\therefore \quad \frac{\lambda_1}{\lambda_2} = \sqrt{\frac{E_2}{E_1}} = \sqrt{\frac{E_1}{4 \times E_1}}$

$= \frac{1}{2}$

$\Rightarrow \quad \boxed{\lambda_2 = 2\lambda_1}$

(20) Answer : (2)
Solution:

$P_a = \frac{1}{f} = \left(1.5 - 1\right)\left[\frac{1}{R_1} - \frac{1}{R_2}\right] = 4$

$\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = 8$

$f_w = \left(\frac{9}{8} - 1\right)\left[\frac{1}{R_1} - \frac{1}{R_2}\right] = \frac{1}{8} \times 8 = 1 \text{ m } = 100 \text{ cm}$ $\Rightarrow$Power = 1 D

(21) Answer : (2)
Solution:
ΔE = 7(4) – [2(1.1)] × 2 = 28 – 4.4 = 23.6 meV

(22) Answer : (2)
Solution:

$u = \frac{1}{2}\varepsilon_0 E^2$

$= \frac{1}{2} \times 8.85 \times 10^{-12} \times 36$

= 1.6 × 10^{-10} J/m^3

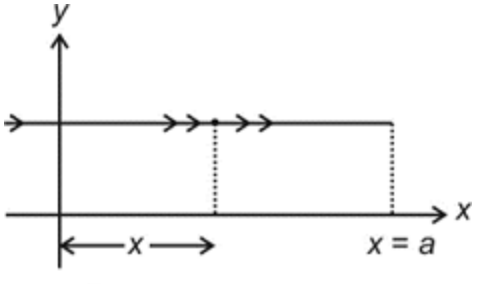
(23) Answer : (1)
Solution:
In a medium, speed of light $v = \frac{c}{\mu}$

$\mu = a + \frac{b}{\lambda^2} \ldots$

$\lambda_{\text{orange}} > \lambda_{\text{blue}}$

$\therefore v_{\text{orange}} > v_{\text{blue}}$

(24) Answer : (2)
Solution:



$v = \frac{c}{\mu}$

$v = \frac{c}{\mu_0\left(1 + \frac{x}{a}\right)}$

Aakash Medical|IIT-JEE|Foundations

$\frac{dx}{dt} = \frac{c}{\mu_0\left(1+\frac{x}{a}\right)}$

$\int_0^t dt = \frac{\mu_0}{C}\int_0^a \left(1+\frac{x}{a}\right) dx$

$t = \frac{3\mu_0 a}{2C}$

(25) **Answer :** (3)
Solution:

L.C = $\frac{\text{Pitch}}{\text{No of divisions on circular scale}}$

Pitch = L.C × Circular scale divisions

= 0.5 $\times 10^{-3}$ × 100 = 5 × 10^{-2} m

= 5 cm

(26) **Answer :** (1)
Solution:

$^A_Z X \rightarrow {}_{Z+1}{}^{A}Y + {}^{0}_{-1}\beta \rightarrow {}^{A-4}_{Z-1}B* + {}^{4}_{2}He \rightarrow {}^{A-4}_{Z-1}B + \gamma$

(27) **Answer :** (3)
Solution:

Use $\beta = \frac{\lambda D}{d}$

$\beta' = \frac{\lambda \times 3D}{\frac{d}{2}} = 6\beta$

(28) **Answer :** (4)
Solution:

de-Broglie wavelength $\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2\,mE}}$

where $E = \frac{1}{2}mv^2$

Squaring both sides,

$\lambda^2 = \frac{h^2}{2mE} \Rightarrow \frac{1}{\lambda^2} =$ (constant) E

Graph passes through origin with constant slope.

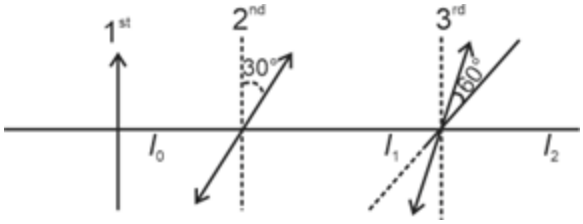
(29) **Answer :** (2)
Solution:

$B_0 = \frac{E_0}{c} = \frac{90}{3\times 10^8} = 3\times 10^{-7}\text{T}$

& $\vec{v} \,||\, \vec{E} \times \vec{B}$

∴Magnetic field is along $\left(-\hat{k}\right)$

(30) **Answer :** (2)
Solution:



$I_1 = I_0\cos^2 30^\circ = I_0 \times \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3I_0}{4}$

$I_2 = I_1\cos^2 60^\circ = \frac{3I_0}{4} \times \left(\frac{1}{2}\right)^2$

$= \frac{3I_0}{16}$

(31) **Answer :** (4)
Solution:

$r_n = r_0 n^2$

$\frac{r_1}{r_3} = \frac{r_0}{r_0 \times 3^2} = \frac{1}{9}$

Aakash Medical|IIT-JEE|Foundations

(32) Answer : (4)
Solution:
For hydrogen atom,

Wave number = $\frac{1}{\lambda} = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

For third line of Paschen series

$n_1 = 3,\ n_2 = 6$

$\frac{1}{\lambda} = 10^7 \left(\frac{1}{9} - \frac{1}{36}\right)$

$= 10^7 \left(\frac{3}{36}\right) = \frac{1}{12} \times 10^7$ m^{-1} = 8.33 × 10^5 m^{-1}

(33) Answer : (4)
Solution:
Total internal reflection is used in optical fibre.

(34) Answer : (2)
Solution:
For same position, where bright fringes due to both wavelengths coincide

$y = \frac{n_1\lambda_1 D}{d} = \frac{n_2\lambda_2 D}{d}$

$n_1\lambda_1 = n_2\lambda_2$

$\frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} \Rightarrow \frac{n_1}{n_2} = \frac{620}{520} = \frac{31}{26}$

Distance of point from central maximum $y = \frac{n_1\lambda_1 D}{d}$

$= \frac{31\times520\times10^{-9}\times2}{0.6\times10^{-3}}$

$= \frac{31\times52\times2}{6} \times 10^{-4}$

$= 537.33 \times 10^{-4} = 53.7 \approx 54$ mm

(35) Answer : (1)
Solution:

$E = \frac{-13.6\times Z^2}{n^2}$, Z = 3 for Li^{2+}

$E_1 = \frac{-13.6\times9}{1} = -122.4$ eV

⇒ K.E. = $-E_1$ = 122.4 eV

(36) Answer : (3)
Solution:

- $r_n = 0.529\ \frac{n^2}{Z}$
- $v_n = 2.19 \times 10^8 \times \frac{Z}{n}$ cm / sec
- $E_n = \frac{-13.6\ Z^2}{n^2}$ eV
- Orbital angular momentum = $\frac{nh}{2\pi}$

(37) Answer : (3)
Solution:

$\frac{I_{max}}{I_{min}} = \left(\frac{\sqrt{I_1}+\sqrt{I_2}}{\sqrt{I_1}-\sqrt{I_2}}\right)^2 = \left(\frac{\sqrt{I}+2\sqrt{I}}{\sqrt{I}-2\sqrt{I}}\right)^2$

$\left(\frac{3}{1}\right)^2 = 9:1$

(38) Answer : (2)
Solution:

$\vec{v}_O = \left(4\hat{i} + 6\hat{j}\right)$ m/s

$\vec{v}_I = \left(-4\hat{i} + 6\hat{j}\right)$ m/s

$\vec{v}_{IO} = \vec{v}_I - \vec{v}_O$

$= \left(-4\hat{i} + 6\hat{j}\right) - \left(4\hat{i} + 6\hat{j}\right)$

$= -8\hat{i}$ m/s

Aakash Medical|IIT-JEE|Foundations

(39) **Answer :** (1)
Solution:
$e = 5 \times 0.01 = 0.05$ cm

(40) **Answer :** (4)
Solution:
According to Einstein's photoelectric equation
$(K.E)_{max} = h\nu - h\nu_0$
$(K.E)_{max} = h(4\nu_0) - h\nu_0$
$(K.E)_{max} = 3h\nu_0$

(41) **Answer :** (4)
Hint:
Energy of nth level in hydrogen atom
$E_n = -\frac{13.6}{n^2}$ eV
Solution:
Energy given to hydrogen atom = 12.75 eV
Energy of electron in excited state = – 13.6 + 12.75 = – 0.85 eV
$= -0.85 = -\frac{13.6}{n^2}$
$n^2 = 16$
$n = 4$
Number of spectral lines in emission
$N = \frac{n(n-1)}{2} = \frac{4\times3}{2} = 6$

(42) **Answer :** (2)
Hint:
In distinct vision mode, magnifying power of compound microscope
$M = \frac{v_0}{u_0}\left(1+\frac{D}{f_e}\right)$ *and* $L = v_0 + u_e$
Solution:
For objective $\frac{1}{v_0} + \frac{1}{2.2} = \frac{1}{2}$
$\Rightarrow \frac{1}{v_0} = \frac{1}{2} - \frac{10}{22} = \frac{11-10}{22}$
$\Rightarrow v_0 = 22\ cm$
$M = \frac{v_0}{u_0}\left(1+\frac{D}{f_e}\right) = \frac{22}{2.2}\left(1+\frac{25}{5}\right)$
= 10 × 6 = 60
Distance of intermediate image from the eyepiece, when final image is at least distance of distinct vision,
$u_e = \frac{f_e D}{D+f_e} = \frac{5\times25}{5+25} = \frac{125}{30}$
$\approx 4\ cm$
$L = v_0 + u_e$
$\simeq 22 + 4$
= 26 cm

(43) **Answer :** (4)
Solution:
Distance of second dark $x_1 = \frac{3\lambda D}{2d}$
$x_1 = \frac{3\lambda D}{2d}$
$\Delta x = \frac{4\lambda D}{d} - \frac{3\lambda D}{2d}$
$= \frac{5\lambda D}{2d}$
$= \frac{5}{2} \times \frac{x}{4}$
$= \frac{5x}{8}$

(44) **Answer :** (4)
Solution:
Wave speed:
$v = \frac{c}{\sqrt{\varepsilon_r \mu_r}} = \frac{3\times10^8}{\sqrt{36}} = 5\times10^7$ m/s
Relation:

Aakash Medical|IIT-JEE|Foundations

$E_0 = vB_0 \Rightarrow B_0 = \frac{E_0}{v} = \frac{90}{5\times10^7} = 1.8\times10^{-6}$ T

(45) Answer : (2)
Solution:
$\Delta E = \frac{hc}{\lambda}$
$\lambda = \frac{hc}{\Delta E} = \frac{hc}{\frac{3}{2}E} = \frac{2hc}{3E}$

CHEMISTRY

(46) Answer : (3)
Solution:
$\pi = \frac{5\times10}{342}\times0.0821\times323$
= 3.88 atm

(47) Answer : (2)
Solution:
Higher the value of K_H, lower is the solubility.
So, correct order of solubility is He < N_2 < H_2 < O_2

(48) Answer : (3)
Solution:
Let molar mass of solute be x g/mol
$\frac{P^0-P_S}{P^0} = \frac{n_B}{n_A+n_B}$
$\frac{10}{100} = \frac{\frac{10}{x}}{\frac{10}{x}+1}$
$10\left(\frac{10}{x}+1\right) = \frac{10\times100}{x}$
$\frac{10}{x}+1 = \frac{100}{x}$
$1 = \frac{100}{x}-\frac{10}{x}$
$1 = \frac{90}{x}$
$x = 90$

(49) Answer : (3)
Solution:
Calculated total pressure P_T
= $X_xP_x^o + X_yP_y^o$
= $\frac{5}{15}\times63 + \frac{10}{15}\times78$
= 21 + 52 = 73 torr
As observed P_T is greater than calculated, P_T
• It shows positive deviation.
• A–B interaction is smaller than A–A and B–B interaction.
• Mixing is endothermic.

(50) Answer : (4)
Solution:
$\Delta T_f = iK_fm = 0.1\times5\ K_f = 0.5\ K_f\ (K_4[Fe(CN)]_6)$
$\Delta T_f = iK_fm = 0.2\times1\ K_f = 0.2\ K_f\ (C_6H_{12}O_6)$
$\Delta T_f = iK_fm = 0.1\times5\ K_f = 0.5\ K_f\ (Al_2(SO_4)_3)$
$\Delta T_f = iK_fm = 0.3\times2\ K_f = 0.6\ K_f(KCl)$

(51) Answer : (3)
Solution:
Molar conductivity increases on dilution of electrolyte solution.

(52) Answer : (4)
Solution:

Aakash Medical|IIT-JEE|Foundations

$R = \frac{\rho l}{A}$

or $R = \frac{l}{kA}$

$\therefore R \propto \frac{1}{k}$

So, the order of resistance is B > C > A > D

(53) Answer : (3)
Solution:
During electrolysis of aqueous sodium chloride using inert electrode, Cl_2 gas is evolved at anode.

(54) Answer : (1)
Hint:
$2H^+(aq) + 2e^- \rightarrow H_2(g)$

$E_{H^+/H_2} = E^\circ_{H^+/H_2} - \frac{0.0591}{2}\log\frac{pH_2}{[H^+]^2}$

Solution:
$E_{H^+/H_2} = 0 - \frac{0.0591}{2}\log\frac{1}{(10^{-5})^2}$

$= -\frac{0.0591}{2}\log 10^{10}$

$= -\frac{0.0591}{2} \times 10 = -0.296\ V$

(55) Answer : (2)
Solution:
$Q = it = 1 \times 2 \times 60 \times 60 = 7200$ C
Q = ne
$\therefore n = \frac{7200}{1.6 \times 10^{-19}} = 4.5 \times 10^{22}$ electrons

(56) Answer : (3)
Solution:

$E^o_{cell} = \frac{2.303\ RT}{nF}\log K_c$

(57) Answer : (2)
Solution:
Unit of rate of reaction is independent of the order of reaction.
r unit is mol L^{-1} s^{-1}

(58) Answer : (1)
Solution:

$k = \frac{2.303}{t}\log\frac{[R]_0}{[R]}$

$= \frac{2.303}{50 \times 60}\log\frac{0.2}{0.0125}$

$= \frac{2.303}{3000}\log 16$

$k = \frac{2.303}{3000} \times 4 \times 0.3 = 9.2 \times 10^{-4} s^{-1}$

(59) Answer : (4)
Solution:
The thermal decomposition of HI on gold surface is an example of zero order reaction.

(60) Answer : (2)
Solution:

From the units of rate constant we can say that it is a zeroth order reaction.

For zeroth order reaction

$A_0 - A = kt$ ... (1)

Now, $A_0 = [OH^-]_i = \frac{5 \times 10^{-6} \times 1000}{0.4}$

= $\frac{5}{4} \times 10^{-2} = 1.25 \times 10^{-2}\ molar$

[A] = 0 (As per the question statement)

$\therefore$ From equation (1)

$1.25 \times 10^{-2} - 0 = 10^6 \times t$

$t = 1.25 \times 10^{-8}$ sec

Aakash Medical|IIT-JEE|Foundations

(61) **Answer :** (3)
Hint:
Rate = $PZ_{AB}\, e^{-E_a/RT}$
Solution:
$e^{-E_a/RT}$ represents the fraction of molecules with energies equal to or greater than E_a.

(62) **Answer :** (2)
Solution:
$[Co(NH_3)_6]Cl_3$ gives 3 mole of Cl^- ions. So, it will consume more equivalents of aqueous solution of $AgNO_3$.

(63) **Answer :** (2)
Solution:
$$4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \rightarrow \underset{(X)}{8Na_2CrO_4} + 2Fe_2O_3 + 8CO_2$$

$$2Na_2CrO_4 + 2H^+ \rightarrow \underset{(Y)}{Na_2Cr_2O_7} + 2Na^+ + H_2O$$

(64) **Answer :** (3)
Solution:

Element	Electron gain enthalpy (kJ mol^{-1})
He	48
Ar	96

(65) **Answer :** (2)
Solution:
Both rhombic and monoclinic sulphur are soluble in CS_2.

(66) **Answer :** (2)
Solution:

$$2Pb(NO_3)_2 \xrightarrow{673\,K} 4NO_2 + 2PbO + O_2$$

(67) **Answer :** (1)
Solution:
$[Ni(CO)_4]$ and $[ZnCl_4]^{2-}$ are diamagnetic and
$[Fe(CN)_6]^{3-}$, $[FeCl_6]^{4-}$ and $[TiCl_3(H_2O)_3]$ are paramagnetic.

(68) **Answer :** (1)
Solution:

$$(NH_4)_2\, Cr_2\, O_7 \xrightarrow{\Delta} N_2 + 4H_2O + Cr_2\, O_3$$

(69) **Answer :** (3)
Solution:
$$8\, MnO_4^- + 3S_2O_3^{2-} + H_2O \rightarrow 8\, MnO_2 + 6\, SO_4^{2-} + 2\, OH^-$$
2 moles of $S_2O_3^{2-} = \frac{8\times 2}{3}$ moles of MnO_4^-

(70) **Answer :** (4)
Solution:

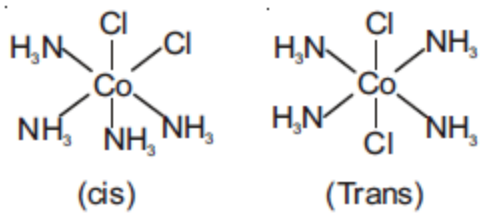
Ion	**Ionic radii (pm)**
Nd^{3+}	99
Eu^{3+}	95
Dy^{3+}	91
Yb^{3+}	86

(71) **Answer :** (3)
Solution:
$V(CO)_6 + Na \rightarrow [V(CO)_6]^- Na^+$
EAN = 35 EAN = 36

Aakash Medical|IIT-JEE|Foundations

EAN = 36 gives extra stability to the complex.

(72) **Answer :** (2)
Solution:



(73) **Answer :** (3)
Solution:
Down the group covalent character decreases.

(74) **Answer :** (3)
Solution:
Ce^{4+} and Yb^{2+} acquire $[Xe]4f^0$ and $[Xe]4f^{14}$ electronic configurations, respectively.

(75) **Answer :** (4)
Solution:

Deacon's process : $4HCl + O_2 \xrightarrow[723\,K]{CuCl_2} 2\,Cl_2 + 2H_2O$

(76) **Answer :** (1)
Solution:
Mn_2O_7 → Acidic

ZnO, Al_2O_3 } Amphoteric

CrO → Basic

(77) **Answer :** (2)
Solution:
XeF_4 → sp^3d^2, square planar

$XeOF_4$ → sp^3d^2, square pyramidal

XeO_2F_2 → sp^3d, see-saw

XeO_3 → sp^3, pyramidal

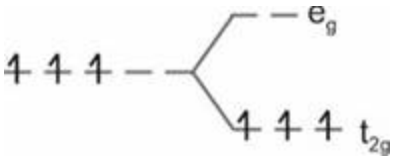
(78) **Answer :** (1)
Hint:
Interstitial compounds are chemically inert.
Solution:
Interstitial compounds retain metallic conductivity.

(79) **Answer :** (4)
Solution:
Excess of copper and iron are removed by the chelating ligands D-penicillamine and desferrioxime B via formation of coordination compounds.

(80) **Answer :** (3)
Solution:
Hypophosphoric acid contains four P–OH bonds.

(81) **Answer :** (2)
Solution:
In $[Cr(H_2O)_6]^{3+}$
$Cr^{3+} \Rightarrow [Ar]4s^0 3d^3$

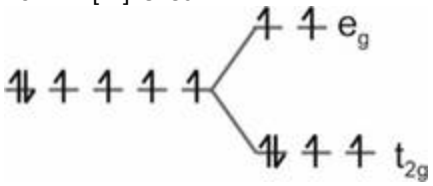
Aakash Medical|IIT-JEE|Foundations



CFSE = $-0.4\Delta_0 \times 3 = -1.2\ \Delta_0$

In $[Fe(H_2O)_6]^{2+}$

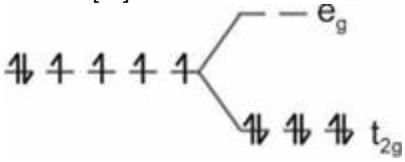
$Fe^{2+} \Rightarrow [Ar]4s^0 3d^6$



CFSE = $-0.4\Delta_0 \times 4 + 0.6\Delta_0 \times 2 = -0.4\Delta_0$

In $[Fe(CN)_6]^{4-}$

$Fe^{2+} \Rightarrow [Ar]4s^0 3d^6$



CFSE = $-0.4\Delta_0 \times 6 \Rightarrow -2.4\Delta_0$

(82) **Answer :** (1)
Solution:
The order of the strength of ligands in the spectrochemical series is
$H_2O < NH_3 < en$
Hence, the wavelength absorbed will be in the order
$[Co(H_2O)_6]^{3+} > [Co(NH_3)_6]^{3+} > [Co(en)_3]^{3+}$

(83) **Answer :** (4)
Solution:
$$(CH_3COO)_2Pb + H_2S \rightarrow \underset{\text{(Black)}}{PbS} + CH_3COOH$$

(84) **Answer :** (1)
Solution:
$H^+ + OH^- \rightarrow H_2O$; $\Delta_r H$ = –57.1 kJ/mol

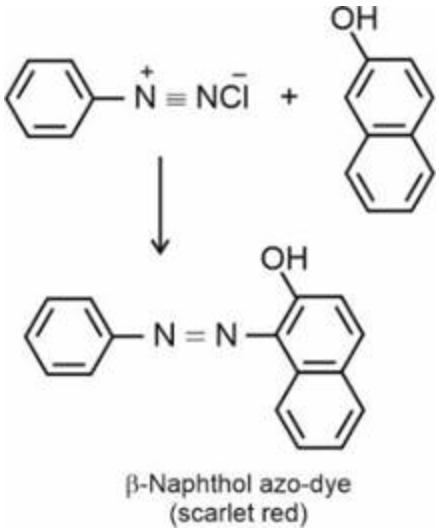
Enthalpy of ionisation = (–x + 57.1) kJ/mol

(85) **Answer :** (1)
Solution:
Group II: Cu^{2+}
Reagent: H_2S gas in presence of dil. HCl.

(86) **Answer :** (3)
Solution:
$ZnCl_2$ and conc. HCl is known as Lucas reagent.
3° alcohol gives immediate turbidity.

(87) **Answer :** (1)
Solution:
$$6C_6H_5OH + FeCl_3 \rightarrow \underset{\text{Violet complex}}{[Fe(C_6H_5O)_6]^{3-}} + 3HCl + 3H^+$$

Aakash Medical|IIT-JEE|Foundations



β-Naphthol azo-dye
(scarlet red)

(88) **Answer :** (2)
Solution:
$[Fe(SCN)]^{2+}$ is blood red in colour.

(89) **Answer :** (1)
Solution:
$(NH_4)_2[Ce(NO_3)_6] + 3ROH \rightarrow [Ce(NO_3)_4(ROH)_3] + 2NH_4NO_3$

(90) **Answer :** (3)
Solution:
Acetylation of aniline is carried out in the presence of strong base than aniline like pyridine, which removes HCl so formed and shifts the equilibrium towards forward direction.

Aakash Medical|IIT-JEE|Foundations

BOTANY

(91) **Answer :** (4)
Solution:
Gametophytic structures are haploid.
Female gametophyte or embryo sac is haploid.

(92) **Answer :** (2)
Solution:
In monoecious plants, male and female flowers are present on the same plant body.

(93) **Answer :** (2)
Solution:
After fertilisation, antipodal cells degenerate.

(94) **Answer :** (4)
Hint:
A typical angiosperm anther is bilobed and dithecous.
Solution:
The proximal end of the filament of stamen are attached to the thalamus or petal of the flower. A typical angiosperm anther is bilobed with each lobe having two theca.

(95) **Answer :** (2)
Solution:
Generative cell is smaller than vegetative cell in pollen grain.
Vegetative cell has large irregularly shaped nucleus.

(96) **Answer :** (2)
Solution:
Egg apparatus contains 3 cells; 1-egg cell, 2- synergids

(97) **Answer :** (4)
Solution:

In flowering plants, double fertilization occurs in embryo sac.

(98) Answer : (1)
Solution:
Since reproductive processes, such as pollination and fertilisation are usually independent of water in angiosperms, seed formation is more dependable.

(99) Answer : (4)
Solution:
An event unique to angiosperms is triple fusion.

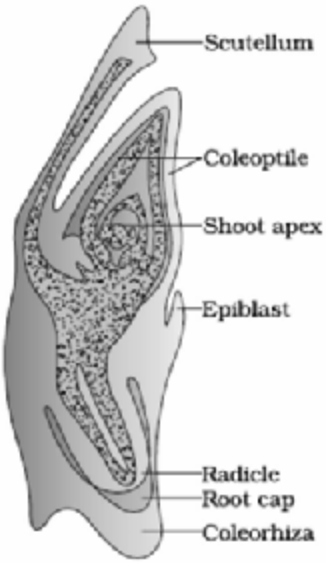
(100) Answer : (4)
Solution:
During double fertilisation one male gamete fuses with diploid secondary nucleus.
Synergids and antipodals degenerate after fertilisation.

(101) Answer : (4)
Solution:
Parthenocarpic fruits develop without the process of fertilization.
Dehydration and dormancy, help in storage of seeds.

(102) Answer : (2)
Solution:
MMC undergoes reductional division to form a haploid cell.
Nucellus undergoes dedifferentiation to form MMC by forming archesporial cell.

(103) Answer : (3)
Hint:
Nectaries attract insects and other animals.
Solution:
Wind pollinated flowers, generally have flowers packed in inflorescence, well exposed stamens, non-sticky pollen and feathery stigma.

(104) Answer : (1)
Solution:



P - Epiblast - Remains of second cotyledon
Q - Coleoptile - Hollow foliar structure, enclosing shoot apex and few leaf primordia
R - Root cap - Covers the terminal of hypocotyl
S - Scutellum - Cotyledon situated laterally to the embryonal axis

(105) Answer : (4)
Hint:
Aquatic plants can be pollinated by air or insects.
Solution:
In *Eichhornia*, the flowers emerge above the level of water and are pollinated by insects or wind.

(106) Answer : (4)
Solution:

Aakash Medical|IIT-JEE|Foundations

Zoological parks, botanical gardens and wild life safari are *ex-situ* conservation strategies.

(107) Answer : (2)
Solution:
Loss of biodiversity in a region may leads to decline in plant production, lowered resistance to environmental perturbations, increased variability in certain ecosystem processes.

(108) Answer : (1)
Solution:
Initially, 25 biodiversity hot spots were identified globally. Subsequently, 9 more were added bringing the number to 34.

(109) Answer : (3)
Solution:
Insects in invertebrates and fishes in vertebrates show maximum biodiversity.

(110) Answer : (2)
Solution:
Biodiversity hotspots' are the regions with very high levels of species richness and high degree of endemism (that is, species confined to that region and not found anywhere else). These hotspots are also regions of accelerated habitat loss. Three of these hotspots – Western Ghats and Sri Lanka, Indo-Burma and Himalaya – cover our country's exceptionally high biodiversity regions.'

(111) Answer : (2)
Solution:
Presently 23% of all mammal species world-wide are facing threat of extinction.

(112) Answer : (4)
Solution:
Bastar district was earlier in Madhya Pradesh but now comes in Chhattisgarh.

(113) Answer : (2)
Solution:
The value of Z lies between 0.6 to 1.2 for a large area like entire continent. If the value of Z is more, the slope will be more steeper.

(114) Answer : (3)
Solution:
Although India has only 2.4 per cent of the world's land area, its share of the global species diversity is an impressive 8.1 per cent.

(115) Answer : (2)
Hint:
Stellar's sea cow got recently extinct from Russia.
Solution:
Dodo got extinct from Mauritius, Thylacine from Australia and Quagga got extinct from Africa.

(116) Answer : (1)
Hint:
Eichhornia is an aquatic plant.
Solution:
Eichhornia was introduced by Europeans in India. It clogges water bodies and resulting in death of several aquatic plants and animals.

(117) Answer : (2)
Hint:
A stable community shows low variability of ecosystem processes like productivity, water use, pest and disease cycle.
Solution:
A stable community does not show too much variation in productivity from year to year.

(118) Answer : (3)
Solution:
The historical 'Convention on Biological Diversity' known as 'Earth Summit' held in Rio-de-Janeiro in 1992.

(119) Answer : (1)
Solution:
1. Experimental verification of chromosomal theory of inheritance was done by T.H Morgan.
2. Hugo de Vries, Tschermak and Correns independently rediscovered Mendel's work on inheritance of characters.

(120) Answer : (2)
Hint:

Aakash Medical|IIT-JEE|Foundations

This pedigree shows inheritance of an X-linked recessive disorder.

Solution:

Female I(R) is heterozygous for the disorder.

Genotype of II(P) can be XX or XX^C

whereas genotype of II(Q) will be X^CX.

(121) Answer : (3)

Solution:

A recessive character that was not expressed in heterozygous condition may be expressed again when it becomes homozygous. Hence, characters segregate while formation of gametes.

(122) Answer : (1)

Solution:

Dominance is not an autonomous feature of a gene.

F_2 dihybrid test cross ratio would not be 1 : 1 : 1 : 1, because no recombinant will be obtained. Only parental types are obtained *i.e.* ratio will be 1 : 1.

(123) Answer : (3)

Solution:

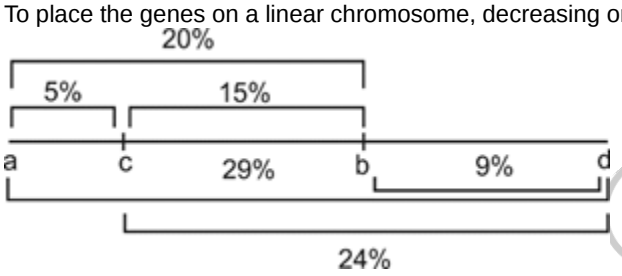
Out of 16 seeds obtained by selfing, one would be wrinkled-green. Therefore, out of 480 seeds, 30 would by wrinkled-green.

(124) Answer : (2)

Solution:

1% recombination frequency = 1 centi Morgan

To place the genes on a linear chromosome, decreasing order of recombination frequency will be considered.



(125) Answer : (2)

Solution:

Factors occur in pairs in diploid organisms. Traits of a character present on the homologous chromosomes, get separated during gamete formation.

(126) Answer : (2)

Solution:

Drosophila can grow easily on synthetic medium, its life cycle is of two weeks and male and female flies are easily distinguishable.

(127) Answer : (4)

Solution:

The female bird has heteromorphic sex chromosome as the females are heterogametic. The female individual produces two different types of gametes. Thus, the egg determines the sex of offspring.

(128) Answer : (3)

Hint:

An individual affected with Turner's syndrome will be a sterile female.

Solution:

Turner's syndrome develops due to monosomic condition where a normal sperm (22 + X) fuses with an abnormal egg (22 + 0) or an abnormal sperm (22 + 0) fuses with a normal egg (22 + X)

(129) Answer : (1)

Solution:

For the child having blood group O, both of his parents should have the allele *i*.

(130) Answer : (3)

Solution:

Klinefelter's syndrome is caused due to the presence of an additional copy of X-chromosome.

Down's syndrome is due to trisomy of an autosome.

(131) Answer : (4)

Solution:

Aakash Medical|IIT-JEE|Foundations

Gynaecomastia (development of breast) is seen in individuals affected with Klinefelter's syndrome.

(132) Answer : (4)
Solution:
The genotype rr represents the recessive condition and produces white flowers in *Antirrhinum*.

(133) Answer : (1)
Solution:
Suppose, F_1 hybrid is 'Aa' and a dominant parent is 'AA'. According to the given question,

$$\frac{F_1}{Aa} \times \frac{\text{dominant parent}}{AA}$$
$$\downarrow$$
$$AA, AA, Aa, Aa$$
$$2 : 2$$
$$\Rightarrow 1 : 1$$

(134) Answer : (4)
Solution:
Flower colour in dog flower is an example of incomplete dominance.

(135) Answer : (2)
Hint:
Colourblindness is sex linked recessive disorder.
Solution:

$XY \times X^cX$

$\downarrow$

$XX^c, XX, \underbrace{X^cY, XY}_{\text{Sons}}$

50% chances

Aakash Medical|IIT-JEE|Foundations

ZOOLOGY

(136) Answer : (3)
Solution:
Implant works for 3-5 yrs.

(137) Answer : (1)
Solution:
Darwin's finches represent the phenomenon of adaptive divergence.

(138) Answer : (3)
Solution:
CT uses X-rays to detect cancer of internal organs by generating 3 D image of the internal organs.

(139) Answer : (2)
Hint:
Hormone releasing IUD.
Solution:
Hormone releasing IUDs – Progestasert, LNG – 20, Copper releasing IUDs – CuT, Cu7, Multiload 375.

(140) Answer : (4)
Solution:
No ozone layer was present in primitive atmosphere.

(141) Answer : (3)
Solution:
ZIFT is Zygote Intra Fallopian Transfer.

(142) Answer : (4)

Hint:

Ideal contraceptive

Solution:

An ideal contraceptive does not interfere with the sexual drive/act of the user.
It is usually easily available and user friendly.

(143) Answer : (2)

Solution:

Numbat and wombat are Australian marsupials and they share a common ancestor. Therefore they represent adaptive radiation.

(144) Answer : (3)

Solution:

When HIV (retrovirus) enters the host cell, the enzyme reverse transcriptase produce a viral copy of viral DNA from RNA genome. This viral DNA gets incorporated into host genome.

(145) Answer : (3)

Solution:

$p + q = 1$
where,
p = Frequency of dominant allele = 0.7
q = Frequency of recessive allele = 1 – P = 1 – 0.7 = 0.3
Also,
$p^2 + 2pq + q^2 = 1$
where
p^2 = Frequency of homozygous dominant individuals = 0.49
q^2 = Frequency of homozygous recessive individuals = 0.09
2pq = Frequency of heterozygous individuals
$= 2 \times 0.7 \times 0.3$
$= 0.42$

(146) Answer : (4)

Solution:

Five factors known to affect Hardy-Weinberg equilibrium are – gene flow, genetic drift, mutation, genetic recombination and natural selection.

(147) Answer : (3)

Solution:

Vaccine provides active immunity.

(148) Answer : (4)

Hint:

Acquired immunity develops due to antigenic stimulus.

Solution:

Acquired immunity is pathogen specific and anamnestic secondary response is characterized by prior encounter with antigen resulting in formation of memory cells during clonal selection.

(149) Answer : (4)

Solution:

Sexual reproduction takes place in the gut of female *Anopheles* mosquito.

(150) Answer : (3)

Solution:

Trichomoniasis is a STD caused by *Trichomonas* which is a flagellated protozoan.

(151) Answer : (2)

Solution:

Genetic drift occurs in small isolated populations as frequency of alleles cannot change suddenly in a large population.

(152) Answer : (1)

Solution:

The given theory was given by Darwin in “origin of species”.

(153) Answer : (3)

Solution:

When the same structure develops along different directions, due to adaptations to different needs, it leads to divergent evolution and those structures are said to be homologous.

Aakash Medical|IIT-JEE|Foundations

(154) Answer : (4)

Hint:
Histamine is a potent vasodilator.

Solution:
The exaggerated response of immune system to certain antigens present in atmosphere is called allergy. Eosinophils and IgE antibodies are increased during allergy. Histamine is not used as a drug to prevent allergy.

(155) Answer : (2)

Solution:
Physical barriers like condoms and diaphragms prevent the deposition of sperms into vagina.

(156) Answer : (3)

Hint:
Origin of life occurred in water.

Solution:
Plants evolved prior to animals. Sauropsids were ancestors of thecodonts.

(157) Answer : (3)

Solution:
Evolution is not always from simple form to complex form.

(158) Answer : (2)

Solution:
Myasthenia gravis and rheumatoid arthritis are auto-immune disorders.

(159) Answer : (3)

Solution:
AIDS was first reported in 1981 and in the last twenty-five years or so, it has spread all over the world killing more than 25 million persons.

(160) Answer : (2)

Solution:
HIV belongs to the category of retroviruses.

(161) Answer : (1)

Hint:
Increasing trend of cranial capacity

Solution:
Among the stories of evolution of individual species, the story of evolution of modern man is most interesting and appears to parallel evolution of human brain and language.

(162) Answer : (2)

Hint:
Man-like primates walked in eastern Africa.

Solution:
Around 3-4 mya old, fossils of man-like bones have been discovered in Ethiopia and Tanzania.
Man-like primates walked in eastern Africa. *Australopithecines* probably lived in east African *(*grasslands). The Neanderthal man lived in near east and central Asia. *Ramapithecus* was more man-like and *Dryopithecus* was more ape like.

(163) Answer : (1)

Solution:
Sedimentary rocks are formed by accumulation and lithification of sediments.

(164) Answer : (3)

Hint:
During the post-industrialisation, tree trunks became dark due to industrial smoke.

Solution:
During the post-industrialisation period, the tree trunks became dark due to industrial smoke and soot. Under this condition, the white-winged moths did not survive due to predators; dark winged or melanised moths survived.
In a mixed population, those that can better adapt, survive and increase in population size. Remember that no variant is completely wiped out.

(165) Answer : (2)

Hint:
Interferons are released from virus infected cells.

Solution:
Physiological barriers include acid in stomach, saliva in mouth and tears from eyes. Ruptured RBCs release haemozoin which is responsible for chills and high fever recurring every three to four days.

(166) Answer : (2)

Aakash Medical|IIT-JEE|Foundations

Hint:
Identify a case of Sewall-Wright effect.
Solution:
When gene migration occurs many time, there will be gene flow. If the change in allele frequency occurs by chance, it is called genetic drift. Following two effects are ramifications of genetic drift i.e., (1) Founder's effect (2) Bottleneck effect.

(167) Answer : (4)
Solution:
Use of MTP to get rid of a female foetus is totally against what is legal in India.

(168) Answer : (2)
Solution:
Opioids are obtained from latex of poppy plant.

(169) Answer : (3)
Hint:
Pregnancy has lasted more than 12 weeks
Solution:
According to the MTP (amendment) Act- 2017 a pregnancy may be terminated on certain considered grounds within the first 12 weeks of pregnancy on the opinion of one registered medical practitioner. If the pregnancy has lasted more than 12 weeks, but fewer than 24 weeks, two registered medical practitioners must be of the opinion, formed in good faith, that the required ground exists.

(170) Answer : (1)
Hint:
Action of cyclosporin drug inhibits this immune response.
Solution:
Cell-mediated immune response comprising cytotoxic T cells (killer cells) is chiefly responsible for graft rejection.

(171) Answer : (1)
Solution:
α-interferons are cytokine barriers and aid in destroying tumor.

(172) Answer : (4)
Solution:
Coke is a stimulant.

(173) Answer : (4)
Solution:
Option (d) is a mismatch because bacteria like *Streptococcus pneumoniae* and *Haemophilus influenzae* are responsible for the disease pneumonia in humans which infects the alveoli [air-filled sacs] of the lungs.
Option (a) is not the answer because, amoebiasis is caused by *Entamoeba histolytica*. Its symptoms include constipation, abdominal pain and cramps, stool with excess mucus and blood clots.
Option (b) is not the answer because *Epidermophyton* are responsible for ringworms, which is one of the most common fungal infections in man. Appearance of dry, scaly lessions on various parts of body such as skin, nails and scalp are the main symptoms of this disease.
Option (c) is not the answer because a slowly developing chronic inflammation of lymphatic vessels of the lower limbs are the symptoms of elephantiasis or filariasis which is caused by *Wuchereria* (*W. bancrofti* and *W*. *malayi*).

(174) Answer : (4)
Hint:
H_2L_2 represents 2H and 2L chains.
Solution:
Antibody is tetramer having heavy and light chains, so represented as H_2L_2. Both heavy and light chains are linked together by interchain disulphide bridges.

(175) Answer : (3)
Solution:
Cycads and angiosperms were evolved from seed ferns in Permian period.

(176) Answer : (3)
Solution:
Normal cells show a property called contact inhibition by virtue of which contact with other cells inhibits their uncontrolled growth.

(177) Answer : (1)
Solution:
The correct answer is option (1) as it includes all the correct statements.Saheli is a new oral contraceptive for females, developed by the scientists at Central Drug Research Institute

(CDRI) Lucknow, India.

(178) Answer : (3)

Hint:

Extra terrestrial origin of life.

Solution:

According to Panspermia, life was transferred from one planet to other in the form of small units called spores or seeds or sperms.

According to the theory of biogenesis, life arose from some pre-existing life.

According to the theory of abiogenesis, life originated from non-living matter.

(179) Answer : (1)

Hint:

Lives on both land and water

Solution:

Lobefins evolved into the first amphibians that lived on both land and water. The amphibians evolved into reptiles. Reptiles evolved into aves and mammals.

(180) Answer : (3)

Solution:

The diagram represents tubectomy in which a small part of fallopian tube is tied up through a small incision.

Aakash Medical|IIT-JEE|Foundations